

Finding Files

A typical running RHEL system has a few hundred thousand files distributed across various file systems. Sometimes it is imperative to look for one or more files based on certain criteria. One example would be to find all files owned by employees who left the company over a year ago. Another example would be to search for all the files that have been modified in the past 20 days by a specific user. For such situations RHEL offers a command called *find*. You supply your search criteria and this command gets you the result on the screen or redirect to a file. You can also instruct this utility to execute a command on the files as they are found. This command is expounded in the following sub-section.

Using the find Command

The *find* command recursively searches the directory tree, finds files that match the specified criteria, and optionally performs an action on the files as they are found. This powerful tool can be tailored to look for files in a number of ways. The search criteria may include tracking for files by name, ownership, owning group, permissions, inode number, last access or modification time, size, and file type.



With the *find* command, files that match the specified criteria are located and the full path to each file is displayed on the screen by default. Let's look at a few examples.

To search for a file called *newfile* by its name in *user1*'s home directory (assuming *newfile* exists):

```
$ cd
```

```
$ find . -name newfile -print
```

`-print` is optional. The *find* command, by default, displays the results on the screen. You do not have to specify this option.

To search for files and directories in */dev* that begin with *vg00* followed by any characters. The `-iname` option directs

the command to perform a case-insensitive search.

```
# find /dev -iname vg00*
```

```
/dev/vg00
```

```
/dev/mapper/vg00-home
```

```
/dev/mapper/vg00-root
```

```
/dev/mapper/vg00-swap
```

To find files smaller than 1MB in size in *root*'s home directory:

```
# find ~ -size -1M
```

```
/root/.cache/abrt/applet_dirlist
```

```
/root/.local/share/gnome-settings-daemon/input-sources-converted
```

```
/root/.local/share/folks/relationships.ini
```

```
/root/.local/share/tracker/data/.meta.isrunning
```

```
.....
```

The tilde ~ character represents a user's home directory.

To find files larger than 40MB in size in */usr* directory:

```
# find /usr -size +40M
```

```
/usr/lib/locale/locale-archive
```

```
/usr/lib/jvm/java-1.7.0-openjdk-1.7.0.51-2.4.5.5.el7.x86_64/jre/lib/rt.jar
```

```
/usr/lib64/firefox/libxul.so
```

```
/usr/share/icons/gnome/icon-theme.cache
```

To find files in the entire root file system with ownership set to *user1* and group membership set to any group other than *user1*:

```
# find / -user user1 -not -group user1
```

```
/var/spool/mail/user1
```

To find files in the */etc* directory that were modified more than 2000 days ago:

```
# find /etc -mtime +2000
```

```
/etc/xdg/menus/documentation.menu
```

```
/etc/xdg/menus/server-settings.menu
```

```
/etc/xdg/menus/settings.menu
```

```
/etc/xdg/menus/start-here.menu
```

To find files in the */var* directory that were modified exactly 12 days ago:

```
# find /var -mtime 12
```

```
/var/log/sa/sa14
```

```
/var/log/sa/sar14
```

To search for character device files in the */dev* directory with permissions set to 666:

```
# find /dev -type c -perm 666
```

```
/dev/kvm
```

```
/dev/net/tun
```

```
/dev/fuse
```

```
/dev/ptmx
```

```
/dev/tty
```

```
/dev/urandom
```

```
/dev/random
```

```
/dev/full
```

```
/dev/zero
```

```
/dev/null
```

To search for character device files in the */dev* directory that are world writeable:

```
# find /dev -type c -perm -222
```

```
/dev/kvm
```

```
/dev/net/tun
```

```
/dev/fuse
```

```
/dev/ptmx
```

```
/dev/tty
```

```
/dev/urandom
```

```
/dev/random
```

```
/dev/full
```

```
/dev/zero
```

```
/dev/null
```

To search for *core* files in the entire directory tree and delete them as they are found without prompting for confirmation:

```
# find / -name core -exec rm {} \;
```

The pattern `{ } \;` is part of the syntax and must be defined that way.

There are numerous other options available with the *find* command. Refer to the command's man pages and try some of them out.

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